

Aavishkar: From Moon to Vyom - India's Cosmic Journey

Celebrating **National Space Day 2025**

BY:- SHIVANG AYUSH
[XIIth Science]

Under; **Manoj Kumar Singh**



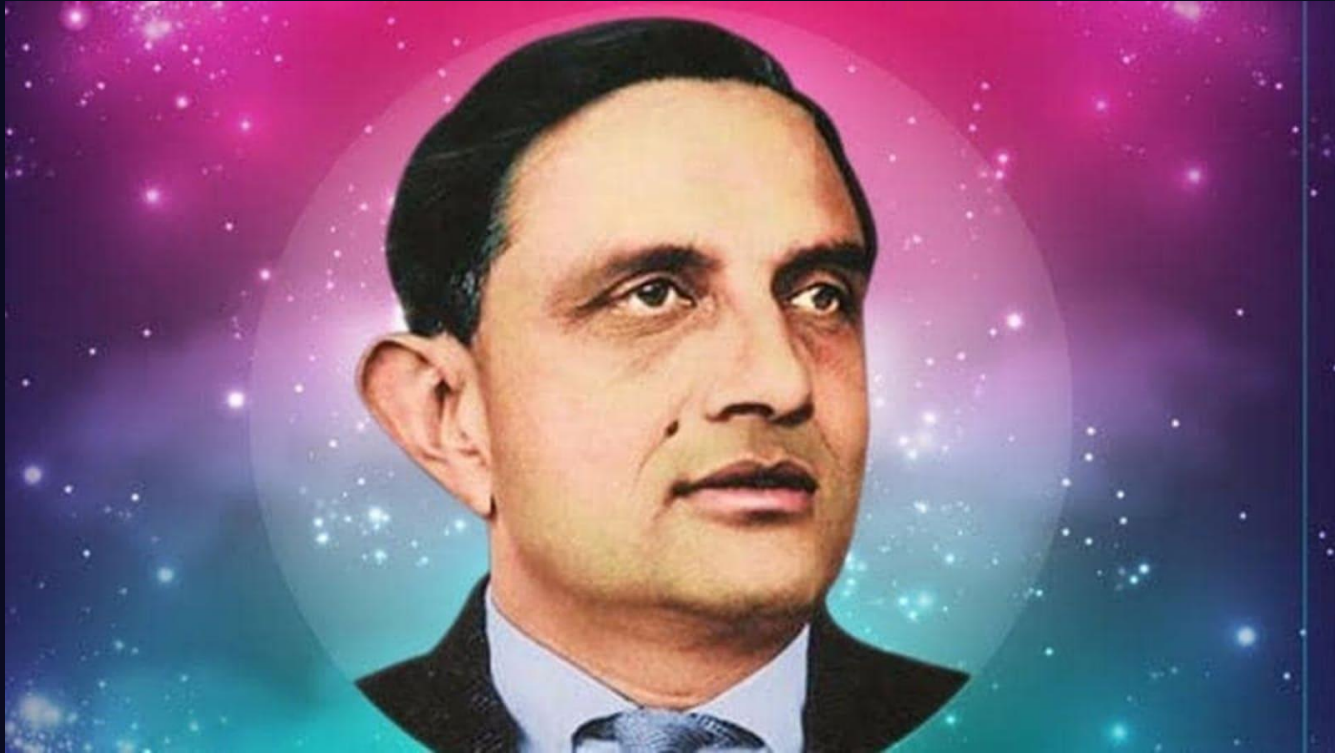
Why August 23rd?

A Date Etched in History

August 23rd marks an extraordinary milestone for India: our historic **first landing on the Lunar South Pole** with Chandrayaan-3. This feat of engineering and scientific prowess led to this day being declared National Space Day, a testament to our nation's relentless pursuit of the cosmos.



Architects of India's Space Dreams



Dr. Vikram Sarabhai



Dr. APJ Abdul Kalam

Our Propulsion Powerhouses

India's Mighty Rocket Fleet



PSLV (The Workhorse)

The Polar Satellite Launch Vehicle, known for its reliability and versatility, has successfully launched numerous Indian and foreign satellites into various orbits, making it a cornerstone of ISRO's missions.



GSLV (The Heavy Lifter)

The Geosynchronous Satellite Launch Vehicle is designed to launch heavier communication satellites into Geosynchronous Transfer Orbit (GTO), expanding India's capabilities in high-orbit missions.



LVM3 (The Bahubali)

The Launch Vehicle Mark-3, ISRO's heaviest and most powerful rocket, is capable of launching 4-ton class satellites to GTO and is designated for future human spaceflight missions like Gaganyaan.

Cosmic Achievements

Missions That Redefined Space Exploration



- **Chandrayaan-1:** India's first lunar probe, which confirmed the presence of water molecules on the Moon.
- **Mangalyaan:** The Mars Orbiter Mission, making India the first Asian nation to reach Martian orbit and the first in the world to do so in its inaugural attempt.
- **Chandrayaan-3:** The historic mission that achieved a soft landing on the Moon's south polar region, a groundbreaking achievement for lunar exploration.

The Gaganyaan Bridge



Gaganyaan is India's first human spaceflight mission aiming to demonstrate the country's capability to send humans into Low Earth Orbit (LEO) and safely return them to Earth.



It involves a crew module designed to carry three astronauts for a short mission (expected to be 3 days at 400 km orbit), followed by a splash-down landing.



Missions are short, designed to demonstrate human spaceflight capabilities and gather initial data on the effects of spaceflight on humans, according to UPSCprep.com.

Meet Vyommitra

Our First Space Companion

Vyommitra, meaning “space friend,” is ISRO’s remarkable **half-humanoid robot** designed to fly on uncrewed test missions before the actual Gaganyaan human spaceflight.

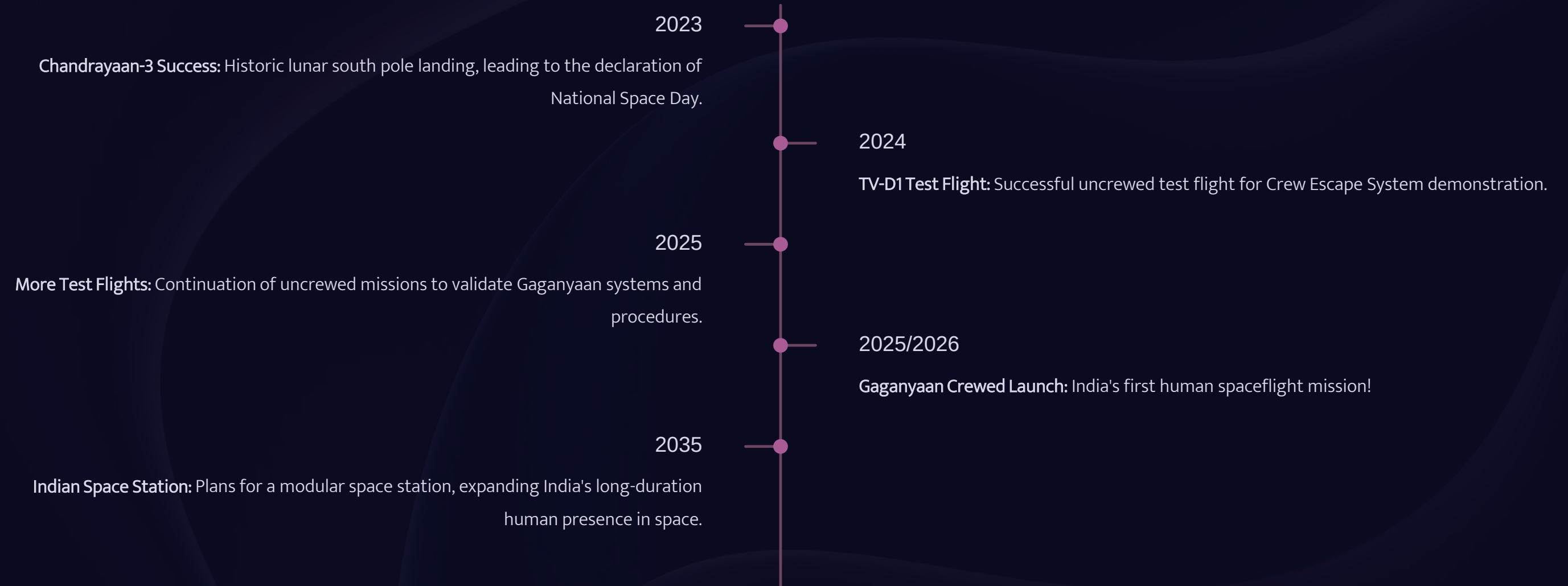
Vyommitra’s role is crucial: it will simulate human functions, monitor module parameters, and provide valuable data on the environment inside the crew module, ensuring future human safety.

This innovative companion helps validate the spacecraft’s systems and conditions for human flight, marking a significant step in our journey to send humans to space.



The Path Ahead

India's Ambitious Space Roadmap



Our journey continues, with each mission building upon the last, paving the way for a future where India plays a leading role in cosmic exploration.

Jai Hind!

Jai Vigyan!

Onward to the stars, united by science and discovery.

